VASILII ANDREEV

POSTGRADUATE RESEARCHER / CIVIL & STRUCTURAL ENGINEER

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RESEARCH PROFILE

Civil and structural engineer specializing in bridge and tunnel engineering. Academic and professional interests integrate inspection and assessment of existing structures, lifecycle management and digital engineering.

Proposed doctoral direction: digital-twin-supported management of existing bridges, linking inspection and structural health monitoring data with engineering and deterioration models to support risk- and cost-informed maintenance.

RESEARCH INTERESTS

Bridge lifecycle management; digital twins for infrastructure; structural health monitoring; condition assessment; remaining-service-life prediction; risk-based maintenance; rehabilitation of existing bridges.

EDUCATION

Postgraduate Studies - Construction Technology and Management

2026 - present

Saint Petersburg State University of Architecture and Civil Engineering (SPbGASU). Research focus: lifecycle management, maintenance planning and digital twins.

Specialist Degree in Civil & Structural Engineering, with Honours

2020 - 2026

SPbGASU; major in Construction of Bridges and Tunnels; six-year programme, 360 ECTS, ISCED Level 7. GPA 4.93/5.00. Thesis: Major Rehabilitation Project for the Alarchin Bridge, a Federal Cultural Heritage Site. Thesis grade: Excellent.

Additional qualifications

2020 - 2025

BIM-platform programming (Peter the Great St. Petersburg Polytechnic University, 2024-2025); Professional Translation in Engineering (SPbGASU, 2020-2022).

RESEARCH EXPERIENCE

Lifecycle Management and Digital Twins for Infrastructure

2023 - present

- Developed an algorithm for assessing effectiveness and performance of lifecycle-management systems for construction assets.
- Extended the assessment methodology to bridge infrastructure and applied it to the Alarchin Bridge case study.
- Investigated the economic role of digital twins in construction lifecycle management and maintenance decision-making.

RESEARCH PUBLICATIONS

Titles below are English renderings of Russian-language publications. Bibliographic details and final issue status should be checked against publishers before formal PhD submissions.

- Andreev, V. K., Kornylev, E. N. Algorithm for Assessing the Effectiveness and Performance of Lifecycle Management of Bridge Structures. Economics of Construction, No. 7, 2026 (final publication details to verify).
- Andreev, V. K., Kornylev, E. N. Application of a Lifecycle-Management Assessment Algorithm: The Alarchin Bridge Case Study. Economics of Construction, No. 8, 2026 (final publication details to verify).
- Andreev, V. K. Digital Twin Technology: Economic Benefits. Questions of Student Science, 1(113), 2026, pp. 481-488.
- Andreev, V. K. Digitalisation of Capital Construction in Russia: Lifecycle Management, Standards and Vulnerabilities. Questions of Student Science, 1(113), 2026, pp. 496-502.
- Andreev, V. K., Nikulin, A. N. Algorithm for Assessing the Effectiveness and Performance of a Facility Lifecycle Management System. High Technologies and Innovations, Belgorod, 2025, pp. 343-349.
- Andreev, V. K. Algorithm for Assessing the Effectiveness of Lifecycle Management of Civil and Industrial Facilities. Modern Building Materials and Technologies, Kaliningrad, 2026.

PROFESSIONAL PRACTICE

URBAN LLC / Structural Design Engineer

2022 - present

- Career progression: Technician, Category III Engineer, Category I Engineer.
- Structural analysis and design of civil and industrial structures; inspections, measurements, defect assessment and verification calculations.
- Design and construction documentation, engineering coordination, repair and strengthening proposals.

RESEARCH AND DESIGN SOFTWARE

SOFiSTiK, SCAD Office, LIRA-SAPR, introductory PLAXIS 2D; AutoCAD, Revit, Tekla Structures, Civil 3D.

LANGUAGES AND ACADEMIC PROFILES

Russian - native; English - B2 self-assessed; German - A2.

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Google Scholar: https://scholar.google.com/citations?user=Lcn4pJkAAAAJ

Portfolio: https://vaasiilii.github.io/

ResearchGate: https://www.researchgate.net/profile/Vasilii-Andreev

Academic and professional references available upon request.